Evaluate yourself 4:

A solution of $[Co(NH_3)_4I_2]Cl$ when treated with $AgNO_3$ gives a white precipitate. What should be the formula of isomer of the dissolved complex that gives yellow precipitate with $AgNO_3$. What are the above isomers called?

Solvate isomers.

The exchange of free solvent molecules such as water , ammonia, alcohol etc.. in the crystal lattice with a ligand in the coordination entity will give different isomers. These type of isomers are called solvate isomers. If the solvent molecule is water, then these isomers are called hydrate isomers. For example, the complex with chemical formula $CrCl_3.6H_2O$ has three hydrate isomers as shown below.

$[Cr(H_2O)_6]Cl_3$	a violet colour compound and gives three chloride ions in solution,
$[Cr(H_2O)_5Cl]Cl_2.H_2O$	a pale green colour compound and gives two chloride ions in solution and,
$[Cr(H_2O)_4Cl_2]Cl.2H_2O$	dark green colour compound and gives one chloride ion in solution

5.5.2 Stereoisomers:

Similar to organic compounds, coordination compounds also exhibit stereoisomerism. The stereoisomers of a coordination compound have the same chemical formula and connectivity between the central metal atom and the ligands. But they differ in the spatial arrangement of ligands in three dimensional space. They can be further classified as geometrical isomers and optical isomers.

Geometrical isomers:

Geometrical isomerism exists in heteroleptic complexes due to different possible three dimensional spatial arrangements of the ligands around the central metal atom. This type of isomerism exists in square planar and octahedral complexes.

In square planar complexes of the form $[MA_2B_2]^{n\pm}$ and $[MA_2BC]^{n\pm}$ (where A, B and C are mono dentate ligands and M is the central metal ion/atom), Similar groups (A or B) present either on same side or on the opposite side of the central metal atom (M) give rise to two different geometrical isomers, and they are called, *cis* and *trans* isomers respectively.

The square planar complex of the type $[M(xy)_2]^{n\pm}$ where xy is a bidentate ligand with two different coordinating atoms also shows *cis-trans* isomerism. Square planar complex of the form $[MABCD]^{n\pm}$ also shows geometrical isomerism. In this case, by considering any one of the ligands (A, B, C or D) as a reference, the rest of the ligands can be arranged in three different ways leading to three geometrical isomers.

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